

Examining product quality and competitiveness via online reviews: An integrated approach of importance performance competitor analysis and Kano model

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ABSTRACT

Understanding and enhancing product quality and competitiveness are vital in today's market. By leveraging the significant potential of online reviews for market-wide consumer research and strategic quality enhancement, this study introduces a novel review analytics approach that combines Importance Performance Competitor Analysis (IPCA) with the Kano model to evaluate and improve product attributes. The proposed Kano-Weighted IPCA (KWIPCA) framework assesses the importance and performance of product attributes by independently weighting positive and negative reviews according to the principles of the Kano model, while also considering multiple competitors in the market. The final output is a user-friendly, four-quadrant matrix that provides comprehensive insights into the market importance of attributes, their relative achievement, and competitors' performance distribution. This KWIPCA matrix enables the evaluation of a product's strengths and weaknesses relative to competitors, identification of critical areas needing immediate improvement, and offers strategic guidance for enhancing overall customer satisfaction. To demonstrate and validate the KWIPCA framework, a case study was conducted using 8567 reviews of 22 five-star hotels in Seoul, South Korea. The results confirm the KWIPCA's effectiveness in providing actionable insights and competitive benchmarking.

1. Introduction

Creating products—whether physical goods or services—that satisfy customers and continuously improve in quality is crucial in today's competitive market. To achieve this, it is essential to understand which aspects of a product consumers consider important and the extent to which these aspects are fulfilled (Yi and Oh, 2022; Zhang and Xu, 2024; Zhang et al., 2024). This knowledge enhances a product or service provider's ability to better meet consumer needs.

Importance-Performance Analysis (IPA), first proposed by Martilla and James (1977), has become a widely adopted tool across many industries for gaining such insights (Tontini, 2016; Kim and Kwak, 2023). In IPA, *importance* refers to how much customers value specific aspects, while *performance* indicates how well the product meets those expectations. By plotting these two dimensions for various attributes on a matrix or graph (see Fig. 1(a)), IPA enables businesses to focus on critical yet underperforming areas. This approach helps pinpoint where improvements are necessary to enhance overall quality and effectively meet customer expectations.

With the assistance of recently advanced natural language processing (NLP) techniques that automatically process textual data, modern IPA practices analyze vast amounts of online customer reviews to retrieve importance and performance metrics (Joung and Kim, 2021; Ma et al., 2024). These NLP-driven approaches not only replace labor-intensive consumer surveys but also enhance the objectivity of the analysis. A typical review-based IPA often incorporates sentiment analysis and counts positive and negative words associated with specific attributes. The importance of these attributes is determined by how frequently they are mentioned in reviews, while performance is assessed by the extent to which positive comments outweigh negative ones.

This study suggests an advanced IPA process that further elicits latent evaluations on importance and performance from review data. Firstly, it expands the scope of analysis to the entire market rather than being constrained to a single offering. Traditional survey-based IPAs often focus solely on one product or service. Although Albayrak (2015) extended it to importance performance competitor analysis (IPCA) which considers primary competitors, adding each competitor to a survey incurs high costs. Additionally, these surveys inevitably impose

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selection bias. In contrast, review comments collected from e-commerce and social platforms provide comprehensive opinions on various competitors, offering a holistic comparison across the entire market landscape and enabling more effective IPCA (Jin et al., 2019a,b; Park et al., 2020; Chen et al., 2024). Leveraging this advantage, the new approach proposed in this study evaluates the importance of individual attributes market-wide and assesses the relative performance and competitiveness of all available products.

Secondly, the proposed approach reveals the distinct impacts of positive and negative impressions (also known as bipolar sentiments) on customer satisfaction and measures them independently for each attribute. Existing review-based IP(C)As typically use the amount or ratio of positive comments as a measure of performance commonly for all attributes. However, the well-known *Kano* quality model (Kano et al., 1984) points out that either positive or negative sentiment may contribute more significantly than its counterpart, and their magnitude is genuine property of each product attribute. Attributes where positive sentiment has a greater impact are classified as *attractive qualities*, while those driven by negative sentiment are classified as *must-be qualities* (see Fig. 1(b)). In light of this, this study introduces a Kano-weighted performance measure that combines ratios of positive and negative comments with their respective weights, which are assessed independently for each attribute. The importance measure is similarly derived from these bipolar weights, following the principles of the Kano quality model. Based on these measures, the proposed IPCA approach is referred to as the Kano-weighted IPCA (KWIPCA).

KWIPCA uncovers true strengths and weaknesses that may be hidden in the nominal counts of positive and negative comments. For example, even if an attribute receives many positive comments and only a few negative ones, it could still be a weakness if the positive sentiments are low-weighted, and the negative ones are high-weighted (a must-be quality). Conversely, an attribute with many negative comments and a few positive ones could be a strength if it is an attractive quality. While some previous studies (Li et al., 2020; Wang et al., 2020; Wu et al., 2018) have also attempted to combine Kano analysis with the IPA framework, they merely add Kano types as an additional dimension of attribute classification, overlooking their impact on performance.

In summary, this study introduces KWIPCA, a novel review-based IPCA approach that evaluates the importance and performance of product attributes across all market competitors. The final output is a single matrix that illustrates attribute importance and the performance distribution of all offerings for each attribute. The matrix not only identifies attributes requiring immediate improvement but also provides strategic guidance for enhancing overall customer satisfaction. To

illustrate and validate KWIPCA, a case study was conducted analyzing 8567 reviews of 22 five-star hotels in Seoul, South Korea, offering a concise overview of this market.

The remainder of this paper is organized as follows. Section 2 reviews relevant studies on review-based approaches to IPA and the Kano model. Section 3 describes the procedure for KWIPCA. Section 4 presents a case study of the Korean hotel service market. Finally, Section 5 outlines conclusions and future research directions.

2. Relevant studies

This section reviews relevant literature on the IPA and Kano analysis using online customer reviews, as summarized in Table 1. While most studies focus on separate analyses of either the IPA or the Kano quality model, a few have explored integrating these approaches, as this study does. It should be noted that this literature review does not cover sentiment analysis techniques, such as attribute extraction and sentiment classification, because the primary focus is on combining review analytics with IPA, the Kano model, and competitor analysis, rather than developing advanced sentiment analysis methods. For further details on review analytics techniques and recent advancements, refer to Khan et al. (2014), Medhat et al. (2014), Hou et al. (2019a), Usama et al. (2020), Hickman et al. (2022), Kaur and Sharma (2023), among others.

2.1. Review-based IPA

The conventional approach to IPA involves conducting customer surveys, a method that can be time-consuming and resource-intensive (Joung and Kim, 2021; Wu et al., 2018). Recently, review-based IPAs have emerged as a more efficient alternative or complement. Table 1 provides a summary of studies on review-based IPA, where product attribute importance and performance levels are assessed through sentiment analysis, culminating in an IPA chart.

In prior studies, attribute importance was commonly measured by *appearance* or *significance*. Appearance refers to the term frequency of each product attribute, representing the proportion of reviews mentioning that specific attribute. In contrast, significance denotes the relative impact of an attribute on overall customer satisfaction. Because attribute significance is implicit, additional approaches such as regression analysis, decision tree classification, or artificial neural network analysis are often used for estimation. For attribute performance evaluation, most studies directly utilized sentiment analysis results. Measures such as the percentage of positive reviews and sentiment scores are frequently employed.

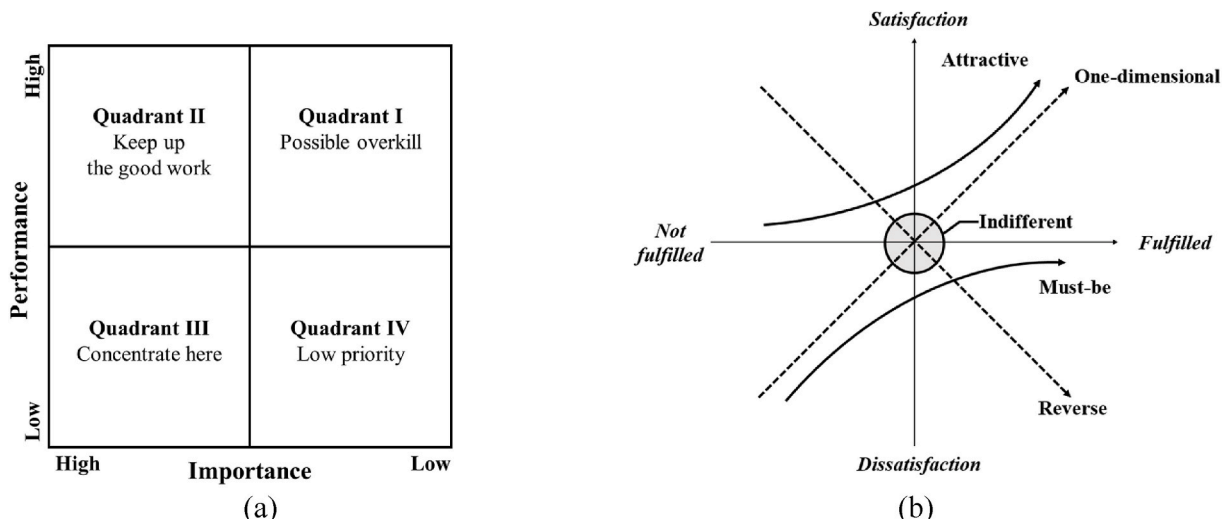


Fig. 1. (a) IPA matrix and (b) Kano model.

Table 1
Summary of review-based studies for IPA and Kano analysis.

Author	IPA	KANO	Competitor analysis	Target
Wang et al. (2016)	Yes		Single competitor	Hotel reservation app
Nam and Lee (2019)	Yes		Several competitors	Airline service
Jeong et al. (2019)	Yes		Single competitor	Smartphone
Bi et al. (2019a)	Yes			Hotel service
Chen et al. (2022)	Yes			Hotel service
Wang et al. (2022)	Yes			Urban park
Liu and Zhang (2020)	Yes			Hotel service
Joung and Kim (2021)	Yes			Smartphone
Kim and Kwak (2023)	Yes			Hotel service
Ma et al. (2024)	Yes			Food delivery service
Shin et al. (2024)	Yes			Hotel service
Xiao et al. (2016)		Yes		Mobile phone
Qi et al. (2016)		Yes		Mobile phone
Bigorra et al. (2019)		Yes		Consumer electronics
Hou et al. (2019b)		Yes	Single competitor	E-reader
Chen et al. (2019)		Yes	Single competitor	Coffee machine
Bi et al. (2019b)		Yes		Mobile phone
Joung and Kim (2022)		Yes		Fitness tracker
Zhou and Yao (2023)		Yes		Tourism spot
Lee et al. (2022)		Yes		Hotel service
Zhang et al. (2023)		Yes		Fresh product
Zhang et al. (2024)		Yes		Mobile phone
Chen et al. (2024)		Yes	Several competitors	Hotel service
Mikulić and Prebežac (2012)	Yes	Yes	Single competitor	Airline service
Wang et al. (2020)	Yes	Yes		Automobiles
Wu et al. (2018)	Yes	Yes	Several competitors	Smartphone
Li et al. (2020)	Yes	Yes		Theme park
This study	Yes	Yes	All competitors	Hotel service

For instance, Wang et al. (2016) conducted a review-based IPA to compare two smartphone applications for hotel reservations: one offered by online travel agencies and the other by hotel brands. They assessed attribute importance and performance based on the proportion of references and positive reviews, respectively. Nam and Lee (2019) employed a review-based IPA combined with Latent Dirichlet Allocation (LDA), which defined product attributes and related words. Attribute importance was determined by the frequency and sentiment intensity of these words, while average star ratings of reviews were used to estimate performance. Similarly, Jeong et al. (2019) performed an IPA using LDA and deep learning-based sentiment analysis. They first defined key topics using LDA, followed by sentiment analysis to determine performance scores. Bi et al. (2019a) also proposed review-based IP(C)As using LDA for attribute identification. They assessed attribute performance based on sentiment strengths from online reviews and estimated attribute importance using an ensemble neural network model.

Chen et al. (2022) carried out a review-based IPA to compare strengths and weaknesses of multiple hotels. Attribute importance was derived from term frequency-inverse document frequency (TF-IDF), while attribute performance relied on sentiment scores. Wang et al. (2022) presented a review-based IPA for urban parks in Beijing to evaluate the most important attributes of the parks and to identify parks in need of immediate renewal. Liu and Zhang (2020) introduced a review-based IPA to analyze online hotel reviews. They determined the attribute importance through a partial correlation analysis between the star rating of the review and the sentiment scores of the attributes. Joung and Kim (2021) determined the importance of smartphone attributes employing a deep neural network. Estimates from multiple optimal neural networks were combined to obtain more precise importance estimates with lower variance. Kim and Kwak (2023) conducted a dynamic IPA, tracking changes of attribute importance and performance

over time. They proposed a review-based control chart combined with dynamic IPA as a new tool for monitoring complaints and prioritizing attributes for quality improvement. Ma et al. (2024) presented a review-based IPA for food delivery service, combining topic modeling and regression for quality measurement. Using advanced BERTopic modeling, they uncovered several topics that had not been studied in previous research, such as corporate social responsibility, marketing strategy, and payment methods. IPA was conducted using frequency-based importance and sentiment score-based correlation values. Shin et al. (2024) proposed a review-based IPA that estimates attribute importance reliably with interpretable machine learning. A case study was conducted to estimate the importance of hotel service features in Singapore.

2.2. Review-based Kano analysis

The Kano model, first introduced by Kano et al. (Kano et al., 1984), categorizes specific attributes based on their perceived quality in relation to customer satisfaction or dissatisfaction (Kano et al., 1984; Lee et al., 2019; Mkpojiogu and Hashim, 2016). While the Kano type itself does not inherently specify an attribute's level of importance (Lee et al., 2019), it helps rank attributes by assessing how their quality affects customer satisfaction or dissatisfaction. In general, product attributes fall into five Kano types: attractive (A), one-dimensional (O), must-be (M), indifferent (I), and reverse (R).

Attractive needs, A, represent unexpected features that exceed customer expectations, leading to high satisfaction when present. However, their absence does not necessarily cause dissatisfaction. One-dimensional needs, O, correspond to the attributes directly affecting customer satisfaction. Improved quality increases satisfaction, while quality falling below expectations leads to dissatisfaction. Must-be needs, M, are basic requirements expected by customers. Fulfilling them may not increase satisfaction, but their absence causes dissatisfaction. The indifferent needs, I, are attributes customers have minimal interest in or are largely indifferent toward in terms of quality. Fulfilling these attributes does not notably impact customer satisfaction. Reverse needs, R, describe attributes where fulfillment results in consumer dissatisfaction, contrasting with one-dimensional needs.

The traditional method of conducting Kano analysis through surveys tends to be expensive and time-consuming. As a result, there is a growing interest in using online customer reviews for Kano analysis. This review-based Kano analysis typically starts with sentiment analysis and involves conjoint analysis to determine the weight of positive and negative sentiments impacting the overall product rating. Regression analysis, decision tree classification, artificial neural network analysis, and similar methodologies are commonly employed for conjoint analysis (Al Rabaie et al., 2021; Lee et al., 2022; Reichenbach et al., 2022; Yang et al., 2022). The Kano type of an attribute is ascertained based on these sentiment weights.

Xiao et al. (2016) proposed a modified ordered choice model and a marginal effect-based Kano model aimed at extracting consumer preferences from online reviews. They integrated the diverse rating behaviors of reviewers into their model and utilized semi-structured reviews—which featured separate sections highlighting the pros and cons—as their input data. Qi et al. (2016) developed a review-based Kano model designed for product improvement. They combined lexicon-based sentiment analysis with classical conjoint analysis for executing Kano analysis. Bigorra et al. (2019) proposed a rule-based classification system for attributes, determining the Kano type based on three criteria: the relative frequency of an attribute's appearance, the proportion of product brands that prominently mention the attribute, and the number of reviews expressing positive or negative sentiments about the attribute. Hou et al. (2019b) analyzed online customer reviews across two successive product generations to track shifts in customer expectations over time. They utilized ordered logistic regression to pinpoint attribute significance and Kano types.

Chen et al. (2019) devised a comprehensive Kano classification framework by integrating multiple artificial intelligence and machine learning techniques, which encompassed sentiment analysis and anomaly detection. Their scheme operated within a three-dimensional space, considering sentiment magnitude in determining Kano type. Bi et al. (2019b) proposed a Kano analysis method employing various machine learning techniques: LDA for attribute extraction, support vector machine for sentiment analysis, and an ensemble neural network for significance analysis. Joung and Kim (2022) proposed an explainable neural network-based approach for the Kano categorization of product attributes from online reviews. A case study of fitness trackers was conducted to validate the increased speed and efficiency of this approach. Zhou and Yao (2023) conducted a review-based Kano analysis employing LDA, lexicon-based sentiment analysis, backpropagation neural network models, and rule-based Kano classification in sequence.

Lee et al. (2022) proposed a review-analytics framework termed importance-Kano analysis (I-Kano analysis). By fusing term frequency and sentiment analyses of online reviews with conjoint analysis, they assessed the impacts of various attributes from three different perspectives that potentially may conflict with each other: appearance, significance, and Kano type. Zhang et al. (2023) also conducted an I-Kano analysis for online retailing of fresh products. LDA and lexicon-based sentiment analysis were applied for the I-Kano analysis. Zhang et al. (2024) presented a review-based Kano model that expanded the scope of analysis to the entire product lifecycle. Sentiment tendency and keywords were analyzed for each life-cycle stage to improve customer satisfaction and pursue market competitiveness. C. Chen et al. (2024) introduced a new method for service design by combining the Kano model and Quality Function Deployment (QFD) using customer reviews. They classified customer requirements through sentiment analysis and translated them into design actions using QFD.

2.3. Integrating IPA and Kano model: research gap and the contributions of this paper

The IPA and Kano models each contribute unique value to improving product quality. IPA specifically identifies attributes that require urgent improvement, especially those considered highly important but showing low performance. In contrast, the Kano model tailors enhancement strategies based on Kano types, acknowledging the potential non-linear relationships between attribute performance and customer satisfaction. As a result, some studies recommend combining review-based IPA and the Kano model to create more effective strategies for quality improvement.

Mikulić and Prebežac (2012) conducted a review-based IPA using a back-propagation neural network to evaluate attribute importance. They focused on the multifaceted nature of attribute importance, assessing it from three potentially inconsistent perspectives: attribute appearance, significance, and Kano type. Attribute performance was directly measured through customer surveys, and the difference in performance compared to a main competitor was considered to identify the company's competitive strengths and weaknesses. Wu et al. (2018) introduced a dynamic IPA method for identifying product and service improvement priorities, where the Kano type of an attribute was used as an index for importance. Lexicon-based sentiment analysis and the C4.5 algorithm were applied for Kano analysis. Li et al. (2020) proposed a Kano-IPA model that retains the traditional measurement of importance used in IPA while adjusting priorities according to Kano classifications. Wang et al. (2020) developed Sentiment-based Importance-Performance Analysis (SIPA), which expands the IPA matrix into a three-dimensional space, representing an attribute's performance alongside its sentiment-based importance, both positive and negative. This space is divided into eight segments by bisecting each dimension at its average value, with each segment suggesting a tailored improvement strategy.

While previous studies have integrated the Kano model only into the importance dimension of IPA, this study extends its application to the

performance dimension by introducing the concept of Kano-weighted performance. Specifically, the proposed KWIPA gives greater weight to positive comments for attractive attributes, while negative comments are weighted more heavily for must-be attributes. This approach retains IPA's intuitive two-dimensional matrix format while incorporating Kano types. Moreover, KWIPA provides a novel presentation of IPA by offering more detailed information about the competitive environment. Instead of comparing the target product to a single competitor or an average value, the relative competitiveness is assessed across the entire distribution of competitors. This comprehensive and intuitive approach enhances the understanding of product quality and helps in developing more effective improvement strategies.

3. Method

This section describes the KWIPCA framework, which consists of five steps: 1) review data preparation and attribute-level sentiment analysis, 2) assessment of the sentiment impact on satisfaction, 3) modeling of attribute importance and Kano types, 4) evaluation of Kano-weighted performance, and 5) formulation of KWIPCA matrix. Fig. 2 provides a technical overview of the KWIPCA framework, and each step is explained in detail in Sections 3.1 through 3.5, respectively. The framework combines widely accepted sentiment and Kano analysis methods for importance scoring, with a novel performance measure and matrix formation technique that offers richer insights. Therefore, the main contribution of this study lies in Steps 4 and 5.

3.1. Step 1: review data preparation and attribute-level sentiment analysis

The KWIPCA framework relies on online review data, which must include textual review comments on a specific product, along with a quantitative rating of overall satisfaction, such as a 5-star rating. Additionally, to establish a reasonable time window for the analysis, each review must be timestamped with its posting date. These are common requirements in review analysis practices.

The primary task in Step 1 is to tag attributes and the sentiment expressed by the reviewer for each attribute in the review text. For this task, the study follows widely used procedures in the literature as outlined below.

As shown in the first three columns of Table 2, review comment j ($j \in \{1 \dots J\}$) is associated with product k ($k \in \{1 \dots K\}$) and satisfaction level S_j ($S_j \in \{1, \dots, 5\}$). The analysis primarily focuses on the correlation between review comments and satisfaction ratings. It is crucial that the data exhibit diversity in both sentiment and satisfaction levels. If all reviews are uniformly positive (or negative), or if all satisfaction ratings are at the same level, further analysis cannot be performed.

The first step is to identify the sentiment—which is positive, negative, or neutral—expressed in a review comment. When a unit of IPA is an attribute of a product or service, the sentiment is also identified for each attribute. If a comment does not tag an attribute, we need to extract the attribute which is mentioned in the review phrase by phrase. As shown in Equations (1) and (2), two binary variables— X_{ij}^{pos} and X_{ij}^{neg} —are employed to denote the sentiment polarity, i.e., satisfaction and dissatisfaction, respectively, toward attribute i .

$$X_{ij}^{pos} = \begin{cases} 1 & \text{attribute } i \text{ is ever mentioned positively in review } j \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

$$X_{ij}^{neg} = \begin{cases} 1 & \text{attribute } i \text{ is ever mentioned negatively in review } j \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

It is important to note that an attribute may exhibit two-sided polarities within a review, denoted as $\{X_{ij}^{pos}, X_{ij}^{neg}\} = \{1, 1\}$, indicating the appearance of both pros and cons in the same review. The reviewer may be satisfied with certain aspects but unsatisfied with others. In cases

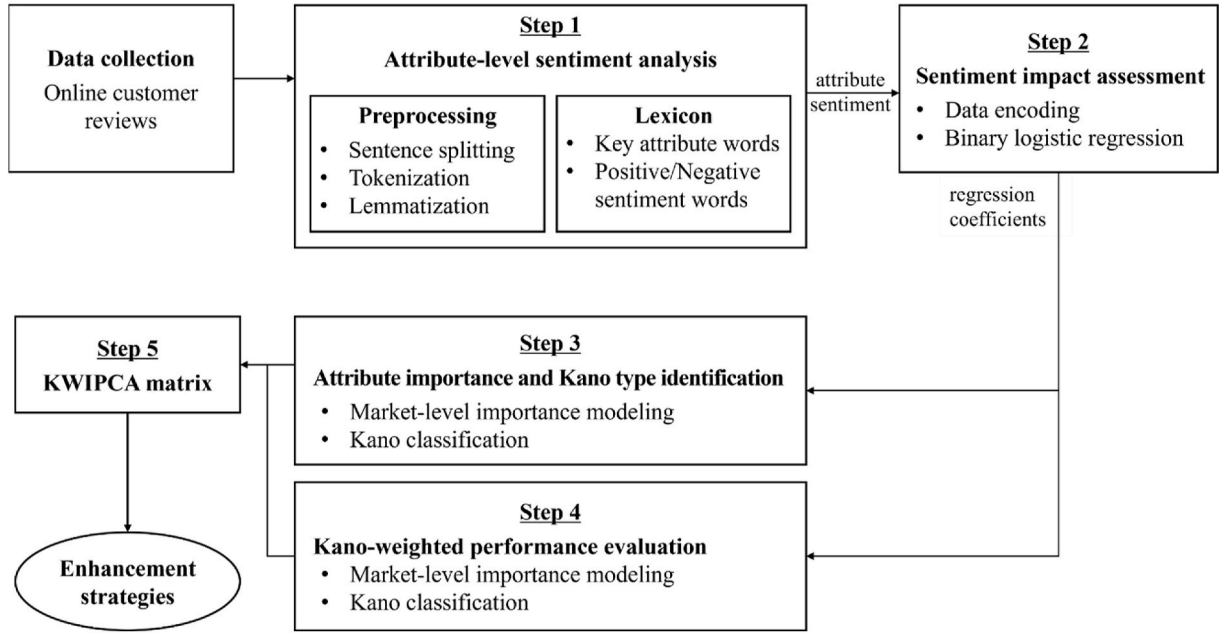


Fig. 2. Overview of the KWIPCA framework.

where attribute i is not mentioned in review j , a neutral sentiment is assumed, setting both X_{ij}^{pos} and X_{ij}^{neg} to 0. This coding scheme has also been employed in previous studies (Jin et al., 2019b). Consequently, the resulting structured data processed through sentiment analysis is exemplified in Table 2.

The proposed framework omits specific details regarding the extraction of attributes from each review phrase and the identification of their sentiment. Various techniques, ranging from lexicon-based matching to sophisticated artificial intelligence classifications (Heng et al., 2018; Kaur and Sharma, 2023; Kumar et al., 2023; Usama et al., 2020), can be used, as long as their output aligns with the proposed coding scheme. In the illustrative example, pre-defined lexicons developed by Lee et al. (2022) and Park et al. (2020) are used to match both attributes and sentiment.

3.2. Step 2: assessment of sentiment impact on satisfaction

In Step 2, the impact of positive and negative sentiments on overall satisfaction is assessed. As discussed in Section 2.2, the review-based Kano literature commonly employs a conjoint analysis approach to determine Kano types. Among various analytical techniques available, ranging from simple correlation to advanced machine learning methods, we adopt a traditional regression approach, consistent with previous studies (e.g., Li et al., 2020; Qi et al., 2016; Wang et al., 2020; Xiao et al., 2016; Lee et al., 2022).

This study specifically employs binary logistic regression to assess the impact of the sentiment on the overall satisfaction. Here, the binary response variable Y_j for review j is defined depending on the overall satisfactions rating S_j . The rating score is provided in the original review typically using a Likert scale ranging from 1 (fully dissatisfied) to 5 (fully

satisfied). Specific notations for each scale vary based on review systems. The response variable Y_j encodes the scale into 0 and 1 with a threshold value γ as shown in Equation (3). In the example in presented in Table 2, γ is set to 4.

$$Y_j = \begin{cases} 0, & S_j < \gamma \\ 1, & S_j \geq \gamma \end{cases} \quad (3)$$

The binary logistic regression derives two regression coefficients per attribute: β_i^{pos} for X_{ij}^{pos} and β_i^{neg} for X_{ij}^{neg} , which demonstrate the impacts of positive and negative sentiments on the overall satisfaction. In KWIPCA, these are utilized in two ways. First, the gap between the two coefficients quantifies the relative importance or significance of the attribute. Second, they serve as criteria for Kano type classification, discussed later in Section 3.3.

Additionally, this study introduces a dummy variable I_{kj} indicating which product k is the target of review j and its coefficient β_k . Since the proposed framework aims to analyze competitive environment, different satisfaction biases toward individual products need to be considered (In Wang et al. (2020), the brand bias is included in the satisfaction regression model for the same purpose.). The coefficient β_k captures this bias, allowing β_i^{pos} and β_i^{neg} more precisely represent the sentiment effects. The regression models without and with a product bias are shown in Equations (4) and (5), respectively.

$$\text{Model I (reference)} : Pr(Y_j = 1) = \frac{\exp\left(\beta_0 + \sum_{i=1}^I \left(\beta_i^{pos} X_{ij}^{pos} + \beta_i^{neg} X_{ij}^{neg}\right)\right)}{1 + \exp\left(\beta_0 + \sum_{i=1}^I \left(\beta_i^{pos} X_{ij}^{pos} + \beta_i^{neg} X_{ij}^{neg}\right)\right)} \quad (4)$$

Table 2
Example of the structured data required for the KWIPCA.

Review j	Product k	Overall satisfaction S_j (1–5)	Attribute 1		Attribute 2		Attribute i		Attribute I	
			X_{1j}^{pos}	X_{1j}^{neg}	X_{2j}^{pos}	X_{2j}^{neg}	X_{ij}^{pos}	X_{ij}^{neg}	X_{Ij}^{pos}	X_{Ij}^{neg}
1	1	4	1	0	0	0	0	0	0	0
2	1	5	1	0	1	0	0	0	0	0
J	K	3	0	0	1	1	1	1	1	0

$$\text{Model II (product bias) : } Pr(Y_j = 1) = \frac{\exp\left(\beta_0 + \sum_{i=1}^I (\beta_i^{pos} X_{ij}^{pos} + \beta_i^{neg} X_{ij}^{neg}) + \sum_{k=2}^K \beta_k I_{kj}\right)}{1 + \exp\left(\beta_0 + \sum_{i=1}^I (\beta_i^{pos} X_{ij}^{pos} + \beta_i^{neg} X_{ij}^{neg}) + \sum_{k=2}^K \beta_k I_{kj}\right)} \quad (5)$$

Since Model II inherently fits the data better due to the inclusion of additional variables, we compare these models using the Akaike information criterion (AIC) and cross-validation measures. The entire dataset is divided into 10 folds of training and validation datasets, while the models are evaluated by deviance R^2 and the area under the receiver operating characteristic curve (AUC) scores for the validation dataset across these folds. The case study in Section 4 illustrates that Model II provides a better fit to the empirical data with respect to both AIC and cross-validated AUCs, as well as goodness of fit.

To select significant variables, we employ the backward elimination procedure. The initial model includes all variables and then excludes the least significant one step by step until all remaining variables have lower p -values than the pre-defined alpha-to-remove threshold, which is set at 0.05 in this study. Finally, the variation inflation factors (VIF) of the variables are calculated to detect multicollinearity. A higher VIF score indicates that the corresponding variable may be dependent on other variables (Kutner et al., 2005).

3.3. Step 3: attribute importance and Kano type identification

Step 3 analyzes an attribute's relative importance and classifies its Kano type based on the sentiment impacts assessed in the previous step. Similar to previous studies that use attribute significance to measure importance (e.g., Qi et al., 2016; Lee et al., 2022), KWIPCA considers an attribute more important when the difference between the impact of

positive sentiment and the impact of negative sentiment on satisfaction is larger. This difference is computed using the coefficients β_i^{pos} and β_i^{neg} and is denoted as gap_i in Equation (6). (This subtraction formula reflects the idea that positive sentiment typically has a stronger effect on increasing satisfaction levels.)

The final importance of attribute i , denoted by IMP_i in Equation (7), is ranked in descending order based on the gap_i values. This is a distinct approach to determining market-level attribute importance. While Li et al. (2020) and Wang et al. (2020) also examined market-level attribute importance based on sentiment impacts, Li et al. (2020) simply matched Kano types to priorities, and Wang et al. (2020) considered positive and negative sentiment impacts independently, without evaluating their combined implications for attribute importance.

$$gap_i = \beta_i^{pos} - \beta_i^{neg} \quad (6)$$

$$IMP_i = \text{Rank of } gap_i \text{ in a descending order} \quad (7)$$

The sentiment coefficients β_i^{pos} and β_i^{neg} also classify Kano types of the attributes. A rule-based classification, adopted from Qi et al. (2016) and Lee et al. (2022), is conducted as depicted in Fig. 3. This classification scheme distinguishes five Kano types: indifferent (I), reversed (R), one-dimensional (O), attractive (A), and must-be (M) qualities. In practice (also in our empirical study in Section 4), most of the attributes fall under the category of one-dimensional qualities. Therefore, this study added two subgroups, namely one-attractive (O/A) and one-must-be (O/M) qualities, to further differentiate them.

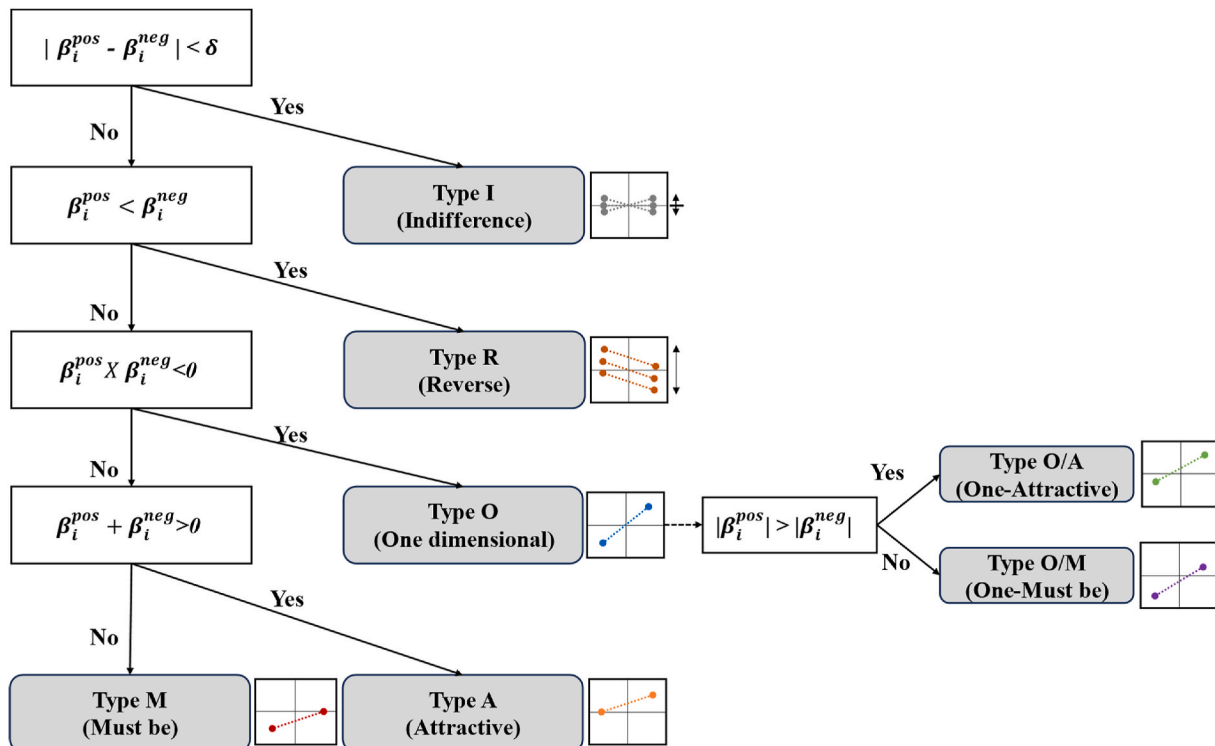


Fig. 3. Rule-based Kano classification in KWIPCA model.

The first branch assesses whether customers are indifferent for an attribute. If values of β_i^{pos} and β_i^{neg} are nearly identical, it suggests that the attribute influences satisfaction simply by being mentioned, irrespective of it being in a positive or negative context. The threshold for this proximity δ is determined by the user, adjustable based on domain knowledge and experience. In this study, an odds ratio of an attribute lying between 0.95 and 1.05 indicates insignificance, setting the threshold at 0.05 (reflected by the absolute value of $\ln(1.05)$ and $\ln(0.95)$). The other branches are fairly straightforward; for a detailed explanation of the rules, please refer to Lee et al. (2022).

3.4. Step 4: evaluation of Kano-weighted performance

While attribute importance is assessed across the entire market, performance is evaluated individually for each product. For each product k , the proportions of positive and negative mentions, denoted as R_{ik}^{pos} and R_{ik}^{neg} , respectively, are calculated as shown in Equations (8) and (9), where $O(k)$ denotes the set of reviews for product k .

$$R_{ik}^{pos} = \frac{\sum_{j \in O(k)} X_{ij}^{pos}}{|O(k)|} \quad (8)$$

$$R_{ik}^{neg} = \frac{\sum_{j \in O(k)} X_{ij}^{neg}}{|O(k)|} \quad (9)$$

In previous studies, various performance measures has been used to evaluate attribute performance, such as the proportion of positive reviews (R_{ik}^{pos}), the proportion of positive reviews relative to all reviews mentioning the same attribute ($R_{ik}^{pos} / (R_{ik}^{pos} + R_{ik}^{neg})$), the ratio of positive to negative reviews ($R_{ik}^{pos} / R_{ik}^{neg}$), and the average star rating of positive reviews ($\sum_{j \in O(k)} S_j \cdot X_{ij}^{pos} / \sum_{j \in O(k)} X_{ij}^{pos}$). However, these measures often overlook the different Kano types of attributes. For instance, negative sentiment toward an attractive-quality attribute has little impact on satisfaction, and positive sentiment toward a must-be-quality attribute is similarly insignificant. More generally, an attribute's satisfaction sensitivity to positive and negative sentiment varies across different Kano types and attributes.

To address this heterogeneity and provide a more accurate assessment of attribute performance, KWIPCA proposes a Kano-weighted performance measure that combines positive appraisals with negative critiques, taking into account the attribute's inherent Kano characteristics. This approach provides a more advanced performance measure than the straightforward counting methods found in the literature. Equation (10) shows the Kano-weighted performance measure, WP_{ik} .

The proportions of positive and negative mentions are weighted by the sentiment coefficients from the logistic regression model. Since these coefficients technically measure the contribution of one positive or negative mention to the probability of the reviewer's overall satisfaction, the weighted score serves as a proxy for the market's average level of satisfaction for product k with respect to attribute i .

$$WP_{ik} = \beta_i^{pos} R_{ik}^{pos} + \beta_i^{neg} R_{ik}^{neg} \quad (10)$$

The performance scores are then normalized using the sample mean m_i and standard deviation s_i across the products, as shown in Equation (11). If PER_{ik} is greater than 0, the performance of product k regarding attribute i is above the market average. Conversely, PER_{ik} less than 0 indicates below-average performance. The absolute value of PER_{ik} indicates the extent of deviation from the average.

$$PER_{ik} = \frac{WP_{ik} - m_i}{s_i} \quad (11)$$

3.5. Step 5: KWIPCA matrix formation

KWIPCA presents a summary of the preceding analysis in a two-dimensional matrix, as shown in Fig. 4. This matrix retains the intuitive format of the traditional IPA while incorporating additional insights, such as Kano-characteristic-adjusted importance, Kano-weighted performance, and the market-level distribution of competitors' performance.

The matrix features importance and performance along the respective axes. Attributes are positioned on the horizontal axis, with attribute i corresponding to the IMP_i -th column. As a result, the most important attribute appears in the leftmost column, followed by others in descending order of importance. The Kano type for each attribute is indicated in parentheses. The vertical axis displays the normalized Kano-weighted performance scores, with the performance score PER_{ik} for product k marked in each attribute's column.

This matrix distinguishes itself from previous IPA and IPCA matrices by presenting the performance distribution of all competitors in the market. As shown in Fig. 4, each attribute column includes a box-plot diagram illustrating the performance score distribution. Additionally, the central horizontal line represents the market average. For example, most competitors perform above average for the first (most important) attribute, though a few perform significantly below average. In contrast, for the fifth attribute, most competitors are below the market average, while a few significantly outperform their peers. This enables analysts to identify whether the market leans toward higher or lower performance

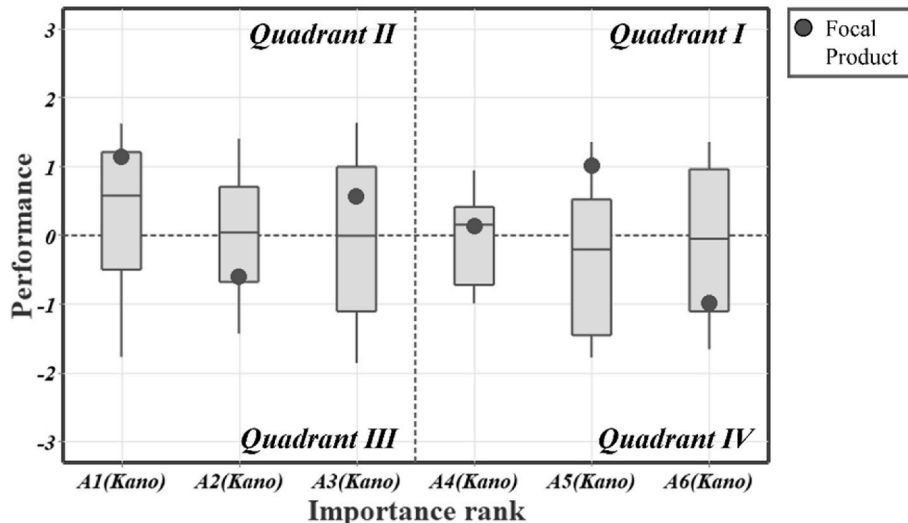


Fig. 4. KWIPCA matrix.

levels for specific attributes.

The figure also shows that the first quartile values are more dispersed than the third quartile values across attributes. In such cases, low-performing competitors deviate significantly from the average for certain attributes, while high-performing competitors tend to deviate uniformly.

This unique visualization of KWIPCA enables an effective evaluation of a focal product's competitive strengths or weaknesses and helps guide the development of appropriate strategies. A quadrant-based strategy is suggested by dividing the matrix into four quadrants, using the central horizontal and vertical lines, which correspond to the average performance and median importance, respectively. When a focal product's attribute falls into each quadrant, the following strategies are recommended.

- **Quadrant I:** Be cautious against overinvestment. The attribute performs well but is less important to customer satisfaction.
- **Quadrant II:** Keep up the good work. The attribute is important to customer satisfaction, and the product outperforms competitors.
- **Quadrant III:** Strive for enhancement. The attribute is critical to customer satisfaction, but customers rate it lower than competitors.
- **Quadrant IV:** Monitor without immediate action. The attribute underperforms, but it is not significant to customer satisfaction.

4. Empirical study

To demonstrate the KWIPCA framework and validate its effectiveness, this section presents a case study analyzing online customer reviews of hotel services in Seoul, South Korea. We conducted web crawling to collect customer reviews from *Tripadvisor*—one of the largest travel websites globally. To limit the analysis to services within the same market segment, we focused solely on reviews of 5-star hotels. All reviews were written in Korean and span the period from January 2012 to June 2020. A total of 8567 reviews from 22 hotels were collected for the case study, with each review text discussing a single hotel without tagging specific attributes. The number of reviews per hotel ranged from 66 to 1245. Overall satisfaction was rated on a 5-point scale: 'Terrible,' 'Poor,' 'Average,' 'Good,' and 'Excellent.'

4.1. Step 1: review data preparation and attribute-level sentiment analysis

In this case study, the product analyzed is a hotel service. Following the framework outlined in Section 3.1, the number of reviews J is 8,567, the number of products K is 22, and $S_j \in \{1, \dots, 5\}$. The first step was defining the product attributes. This study adopted the attribute definitions and associated lexicon from Lee et al. (2022). As shown in Table 3, hotel services are perceived by customers in terms of seven attributes: room condition (RC), service and staff (SS), facilities and conveniences (FC), location and surroundings (LS), dining and food (DF), bathroom condition (BC), and furniture and appliances (FA). Each attribute mentioned in a review sentence was identified by the presence

Table 3
Attribute lexicon (Lee et al., 2022).

Attribute	Associated words ^a (examples)
Room condition (RC)	cleanliness, soundproofing, temperature, view, ...
Service and staff (SS)	staff, receptionist, concierge, internet access, ...
Facilities and conveniences (FC)	atmosphere, lounge, parking, pool, spa, sauna, gym, banquet hall, ...
Location and surroundings (LS)	location, approach, public transportation, landmark, ...
Dining and food (DF)	meal, room service, ...
Bathroom condition (BC)	bathroom, toiletries, ...
Furniture and appliances (FA)	furniture, bed, curtain, appliance, electronics, ...

^a Translated from Korean.

Table 4
Sentiment lexicon (Park et al., 2018).

Sentiment polarity	Associated words ^a (examples)	Number of words
Positive	satisfying, comfortable, nice, kind, friendly, pleased, ...	130
Negative	uncomfortable, noisy, old, unkind, dirty, poor, awful, ...	121

^a Translated from Korean.

of its associated words.

Sentiment for each attribute was determined using a predefined lexicon. This study employed the KNU Korean Sentiment Lexicon (Park et al., 2018), selecting 130 positive and 121 negative words based on their frequency in the collected reviews. Table 4 provides a portion of the sentiment lexicon. When an attribute word appeared together with a positive or negative word in the same sentence, the review was classified as having positive or negative sentiment for that attribute, respectively. According to Lee et al. (2022), the accuracy of lexicon-based sentiment classification is approximately 86%. The analysis produced values for the X_{ij}^{pos} and X_{ij}^{neg} variables, where i is an attribute and j is a hotel service.

4.2. Step 2: assessment of sentiment impact on satisfaction

Binary logistic regression analysis was conducted using 8567 reviews across 22 hotels, with two models represented by Equations (4) and (5): Model I and Model II. The overall satisfaction response of each review was encoded as a binary variable Y_j , with a threshold value $\gamma = 4$. In *Tripadvisor*'s terms, ratings of 'Good' and 'Excellent' were classified as satisfied, whereas all other ratings were considered unsatisfied. Satisfied ratings accounted for 91.3% (7818) of the total reviews. Model I assumes a common bias for all hotels, whereas Model II accounts for heterogeneous biases across hotels. Variables with p-values larger than the alpha removal threshold of 0.05 were eliminated from each model.

Table 5 presents the result of the analysis. The models were validated using a 10-fold cross-validation procedure. Model II, which includes

Table 5
Regression models for the hotel reviews.

Attribute terms	Model I (reference)			Model II ^a (with hotel biases)		
	Coefficient (β_i)	p-value	VIF	Coefficient (β_i)	p-value	VIF
Constant	1.927	0.000		1.483	0.000	
RC pos	0.527	0.000	1.07	0.544	0.000	1.10
RC neg	-1.407	0.000	1.12	-1.401	0.000	1.15
SS pos	1.126	0.000	1.06	0.937	0.000	1.08
SS neg	-1.647	0.000	1.11	-1.631	0.000	1.13
FC pos	0.550	0.000	1.08	0.610	0.000	1.11
FC neg	-0.608	0.000	1.14	-0.516	0.000	1.15
LS pos	—	—	—	—	—	—
LS neg	—	—	—	—	—	—
DF pos	0.820	0.000	1.05	0.788	0.000	1.08
DF neg	-0.696	0.000	1.07	-0.618	0.000	1.08
BC pos	0.470	0.006	1.08	0.447	0.011	1.09
BC neg	-0.971	0.000	1.09	-0.933	0.000	1.10
FA pos	0.697	0.000	1.06	0.697	0.000	1.06
FA neg	-1.043	0.000	1.06	-1.039	0.000	1.06
Model summary						
10-fold deviance R^2	20.90%			24.03%		
10-fold AUC	0.8154			0.8434		
AIC	4015.35			3855.52		
Goodness-of-Fit Test (Hosmer-Lemeshow, p-value)	0.036			0.149		

^a Coefficients for dummy variables (representing 22 individual hotels) are omitted here.

hotel-specific biases, outperformed Model I across all metrics: deviance R^2 , AUC, and AIC scores. It also passed the Hosmer-Lemeshow goodness-of-fit test, while Model I did not. Consequently, Model II was selected for subsequent steps. The attribute-level sentiment coefficients were significant for all attributes except LS. According to VIF scores no notable collinearity was detected between variables.

4.3. Step 3: modeling attribute importance and Kano analysis

The attributes' importance and their Kano types were derived from the sentiment coefficients estimated in the previous step. Table 6 lists the attributes in order of IMP_i , along with the coefficient values and Kano types, excluding LS, which was deemed insignificant. The gap between β_i^{pos} and β_i^{neg} was more than twice as large for the most important attribute (SS) compared to the least important one (FC). This indicates that hotel guests perceive the importance of service attributes quite differently, suggesting that enhancements should be prioritized for the most crucial attributes.

In this case study, all attributes showed positive β_i^{pos} and negative β_i^{neg} values, classifying them as one-dimensional (O) Kano qualities. This suggests that both reducing complaints and exceeding expectations are effective for improving overall satisfaction. More specifically, the attributes FD and FC lean slightly toward surpassing expectations (O/A qualities), while SS, RC, FA, and BC favor reducing complaints (O/M qualities).

4.4. Step 4: evaluation of Kano-weighted performance

The next step was to evaluate the Kano-weighted performance (WP_{ik}) for each hotel k with respect to each attribute i . Table 7 displays the resulting weighted performance WP_{ik} and its normalized score PER_{ik} .

Notably, the weighted performance presented a different picture compared to simply counting positive and negative reviews. For instance, Hotel 5 ranked fourth out of 22 hotels based on the ratio of positive reviews about SS. However, its weighted performance was below the average ($PER_{SS,5} < 0$), as the higher weight assigned to negative reviews reduced its performance score. Conversely, while Hotel 2 ranked eighth in terms of the number of negative reviews regarding DF, its weighted performance score was exceptionally ($PER_{DF,2} = 3.61$), placing it in the top 0.015% of a normal distribution. This is because over 73% of guests reviewing the hotel positively mentioned its dining and food experience, where exceeding expectations is more critical than reducing complaints.

4.5. Step 5: KWIPCA matrix formation

The previous importance and performance scores were summarized in the proposed KWIPCA matrix, as shown in Fig. 5. Attributes were positioned from left to right according to their importance rankings, with box plots illustrating performance distributions across all hotels. Since the performance scores were normalized using the mean and standard deviation, the central horizontal line corresponds to the mean performance.

Where the median value is positive, such as for SS, RC, DF, and BC, more than half of the hotels show above-average performance. For these

attributes, some low-performing hotels drag the average down. Additionally, for attributes like SS, DF, and BC, the range from the first quartile to the median is more spread than from the median to the third quartile. This indicates fiercer competition among high-performing hotels than among low-performing ones. The distribution of DF appears relatively compressed compared to other attributes, despite the normalized scale, due to an extremely high score from Hotel 2, which inflates the standard deviation.

Moreover, the matrix, marked with focal hotels, aids in developing targeted service improvement strategies. Fig. 5 highlights Hotels 9 and 20, selected from the top five hotels most frequently mentioned in the review data. Hotel 9 received the highest overall satisfaction score, while Hotel 20 received the lowest. Moving forward, we will examine the underlying reasons for Hotel 9's high praise and positive evaluations, contrasted with the negative reviews and low evaluations for Hotel 20, to devise tailored enhancement strategies.

First, Hotel 9 placed its two most important attributes, SS and RC, in the second quadrant, whereas Hotel 20 did so for RC and FA. The matrix clearly shows why these two hotels receive contrasting guest evaluations. The most critical determinant is service and staff experience (SS). Hotel 9 excels in providing exceptional service, outperforming all other hotels, while Hotel 20 has the lowest reputation in this area. To enhance their reputation, Hotel 20 should immediately focus on retraining staff. Since SS is an O/M quality type, the priority should be to reduce complaints. Room condition (RC) is excellent for both hotels. While Hotel 20's guests were satisfied with furniture and appliances, this was not sufficient to improve the hotel's overall reputation. In contrast, Hotel 9 has the potential to further enhance its service by updating furniture and appliances.

For less critical attributes—DF, BC, and FC—Hotel 20 performs near average and can continue monitoring these areas without making significant new investments. Hotel 9 shows considerably low performance in DF and FC. Although this low performance suggests that these attributes are not a priority for guests, they could pose a potential risk to the hotel's service quality. It would be beneficial for Hotel 9 to improve the dining and food experience to mitigate this risk. On the other hand, bathroom features (BC), which already receive guest praise, could be deprioritized from a cost perspective, as bathroom condition is categorized as an O/M quality attribute.

5. Discussion

The proposed KWIPCA matrix intuitively visualizes the competitive positioning of market players, enabling each player to scrutinize the reasons behind their high or low reputations among customers for each attribute dimension. To highlight the advantages of KWIPCA over existing methods, we revisit the empirical study involving hotel reviews. For comparison, two additional IPA matrices were constructed: (1) a traditional IPA matrix that scores performance based on the proportion of positive reviews for an attribute relative to all reviews, and (2) a matrix based on Wang et al. (2020) that scores performance based on the proportion of positive reviews for an attribute relative to the total number of reviews specifically mentioning that attribute. Both benchmark matrices lack the Kano weights incorporated by KWIPCA.

Fig. 6 illustrates the benchmark IPA matrices for Hotels 9 and 20, similar to Fig. 5. Notably, the importance dimensions (X-axis) of both matrices use the importance rankings from KWIPCA for easier comparison. The quadrants are split by the median importance rank and average performance score. In the traditional IPA (Fig. 6(a)), average performance is calculated across all attributes of the focal products (Hotels 9 and 20). In contrast, in Wang et al. (2020) (Fig. 6(b)), the average is calculated independently for each attribute across all competing products, resulting in a stepwise splitting line.

For easy comparison of the results, Table 8 shows the quadrant positions of each attribute in the KWIPCA and benchmark matrices. For both Hotels 9 and 20, the two benchmark matrices place many attributes

Table 6
Attributes' importance and Kano types.

Attribute i	IMP_i	gap_i	β_i^{pos}	β_i^{neg}	Kano type
SS	1	2.568	0.937	-1.631	O/M
RC	2	1.945	0.544	-1.401	O/M
FA	3	1.736	0.697	-1.039	O/M
DF	4	1.406	0.788	-0.618	O/A
BC	5	1.380	0.447	-0.933	O/M
FC	6	1.126	0.610	-0.516	O/A

Table 7
Kano-weighted performance of the hotels.

Hotel	SS				RC				FA			
	R_{ik}^{pos}	R_{ik}^{neg}	WP_{ik}	PER_{ik}	R_{ik}^{pos}	R_{ik}^{neg}	WP_{ik}	PER_{ik}	R_{ik}^{pos}	R_{ik}^{neg}	WP_{ik}	PER_{ik}
1	0.545	0.108	0.335	−0.17	0.472	0.176	0.010	−0.46	0.080	0.034	0.020	−0.93
2	0.593	0.066	0.448	0.81	0.168	0.022	0.061	0.12	0.011	0.004	0.004	−1.63
3	0.569	0.036	0.475	1.04	0.563	0.153	0.092	0.48	0.090	0.019	0.043	0.06
4	0.370	0.127	0.140	−1.86	0.475	0.238	−0.074	−1.43	0.055	0.039	−0.002	−1.86
5	0.619	0.114	0.395	0.35	0.585	0.193	0.048	−0.03	0.142	0.028	0.069	1.19
6	0.530	0.136	0.274	−0.70	0.500	0.212	−0.025	−0.86	0.076	0.000	0.053	0.47
7	0.530	0.065	0.391	0.31	0.500	0.330	−0.190	−2.75	0.085	0.020	0.038	−0.14
8	0.655	0.057	0.522	1.44	0.498	0.082	0.157	1.21	0.108	0.011	0.064	0.94
9	0.667	0.050	0.544	1.63	0.487	0.084	0.147	1.10	0.070	0.008	0.040	−0.07
10	0.575	0.080	0.407	0.45	0.333	0.149	−0.028	−0.90	0.115	0.000	0.080	1.64
11	0.434	0.022	0.371	0.14	0.427	0.068	0.137	0.99	0.104	0.022	0.050	0.36
12	0.514	0.076	0.359	0.03	0.459	0.109	0.097	0.53	0.049	0.011	0.023	−0.82
13	0.550	0.077	0.390	0.31	0.666	0.135	0.174	1.41	0.090	0.014	0.049	0.29
14	0.569	0.051	0.449	0.82	0.331	0.090	0.054	0.04	0.135	0.022	0.071	1.24
15	0.439	0.086	0.272	−0.72	0.454	0.111	0.092	0.47	0.139	0.021	0.075	1.41
16	0.300	0.072	0.164	−1.65	0.386	0.143	0.009	−0.48	0.063	0.027	0.016	−1.11
17	0.637	0.104	0.427	0.63	0.568	0.158	0.088	0.43	0.060	0.016	0.025	−0.70
18	0.493	0.036	0.403	0.41	0.406	0.116	0.058	0.09	0.101	0.000	0.071	1.24
19	0.543	0.045	0.435	0.70	0.404	0.088	0.097	0.54	0.080	0.019	0.036	−0.25
20	0.390	0.137	0.142	−1.84	0.571	0.126	0.134	0.95	0.101	0.016	0.054	0.52
21	0.393	0.115	0.180	−1.51	0.483	0.252	−0.090	−1.61	0.073	0.034	0.015	−1.14
22	0.494	0.109	0.285	−0.61	0.432	0.124	0.062	0.14	0.069	0.021	0.026	−0.69

Hotel	DF				BC				FC			
	R_{ik}^{pos}	R_{ik}^{neg}	WP_{ik}	PER_{ik}	R_{ik}^{pos}	R_{ik}^{neg}	WP_{ik}	PER_{ik}	R_{ik}^{pos}	R_{ik}^{neg}	WP_{ik}	PER_{ik}
1	0.426	0.057	0.301	0.95	0.074	0.085	−0.047	−1.78	0.494	0.131	0.234	0.96
2	0.733	0.059	0.541	3.61	0.055	0.011	0.014	0.36	0.176	0.015	0.100	−1.66
3	0.318	0.031	0.231	0.18	0.082	0.033	0.005	0.04	0.377	0.090	0.183	−0.03
4	0.210	0.044	0.138	−0.84	0.072	0.072	−0.035	−1.37	0.331	0.133	0.134	−0.99
5	0.313	0.051	0.215	0.00	0.131	0.057	0.005	0.05	0.523	0.125	0.254	1.36
6	0.242	0.106	0.125	−0.98	0.106	0.106	−0.052	−1.96	0.333	0.212	0.094	−1.77
7	0.315	0.035	0.227	0.13	0.135	0.080	−0.014	−0.65	0.445	0.175	0.181	−0.07
8	0.331	0.040	0.236	0.24	0.127	0.015	0.043	1.36	0.428	0.064	0.228	0.84
9	0.207	0.017	0.153	−0.68	0.079	0.020	0.017	0.45	0.266	0.054	0.134	−0.98
10	0.356	0.046	0.252	0.42	0.069	0.080	−0.044	−1.70	0.414	0.126	0.187	0.05
11	0.247	0.022	0.182	−0.36	0.115	0.025	0.028	0.84	0.502	0.054	0.278	1.82
12	0.341	0.058	0.233	0.20	0.071	0.024	0.009	0.17	0.330	0.102	0.148	−0.71
13	0.188	0.039	0.124	−1.00	0.139	0.046	0.019	0.52	0.450	0.070	0.239	1.05
14	0.399	0.031	0.295	0.89	0.096	0.037	0.009	0.16	0.438	0.094	0.219	0.66
15	0.404	0.086	0.265	0.56	0.150	0.046	0.024	0.69	0.454	0.161	0.194	0.18
16	0.224	0.090	0.121	−1.03	0.045	0.063	−0.039	−1.50	0.390	0.157	0.157	−0.54
17	0.230	0.082	0.131	−0.93	0.148	0.041	0.028	0.84	0.388	0.139	0.165	−0.38
18	0.355	0.065	0.239	0.28	0.116	0.014	0.038	1.21	0.304	0.087	0.141	−0.86
19	0.274	0.029	0.198	−0.18	0.080	0.022	0.015	0.40	0.510	0.072	0.274	1.73
20	0.342	0.075	0.223	0.09	0.132	0.053	0.009	0.18	0.409	0.142	0.176	−0.17
21	0.209	0.060	0.128	−0.96	0.128	0.034	0.025	0.75	0.346	0.145	0.136	−0.95
22	0.247	0.055	0.161	−0.59	0.114	0.021	0.031	0.95	0.430	0.105	0.208	0.46

in different quadrants, leading to inconsistent enhancement strategies. This inconsistency arises from the failure to consider the varying impacts of positive and negative sentiments. In contrast, KWIPCA provides more robust quadrant positions by accounting for these differences. Let us delve deeper into the benchmark matrices to understand how this is achieved.

As the traditional IPA determines performance based on the proportion of positive reviews, it fails to account for negative feedback, potentially leading to a misjudgment of performance. The results for Hotel 9 are largely consistent between the traditional IPA and KWIPCA, as Hotel 9 rarely receives negative comments across all attributes. However, discrepancies emerge with Hotel 20. While SS and FC appear to perform well relative to other attributes in the traditional IPA, KWIPCA reveals that SS actually ranks lowest in the market, and FC performs below average. The traditional IPA overlooks the significant number of negative comments associated with these attributes.

Another issue with the traditional IPA is that the proportion of positive reviews is influenced by attribute appearance (term frequency). A low proportion of positive reviews does not necessarily indicate low satisfaction; it may simply result from low attribute appearance. For

instance, attributes FA and BC for both hotels appear to significantly underperform in the traditional IPA, while KWIPCA rates them above or near the average. This discrepancy is due to their low proportion of positive reviews. However, as Table 7 shows, this low proportion of FA and BC is a market-wide phenomenon, suggesting that the main reason is their low appearance, not low satisfaction—therefore not a cause for concern. Furthermore, as these attributes are more akin to must-be qualities, the infrequency of both positive and negative comments further minimizes their impact.

For instance, the proportions of positive reviews for DF and FC in Hotel 9 are 0.207 and 0.266, respectively, while the averages for competitors are 0.319 (DF) and 0.403 (FC). Fortunately, DF and FC received relatively few negative reviews—0.017 and 0.054, respectively, compared to competitors’ averages of 0.055 (DF) and 0.114 (FC). KWIPCA identifies DF and FC as low-performing attributes by considering both competitors’ performance and the Kano type of attributes. Although Hotel 9 has a lower proportion of negative reviews compared to competitors, it falls behind in the proportion of positive reviews. KWIPCA balances these conflicting results using Kano-weighted performance measures. For O/A type attributes like DF and FC, a lower

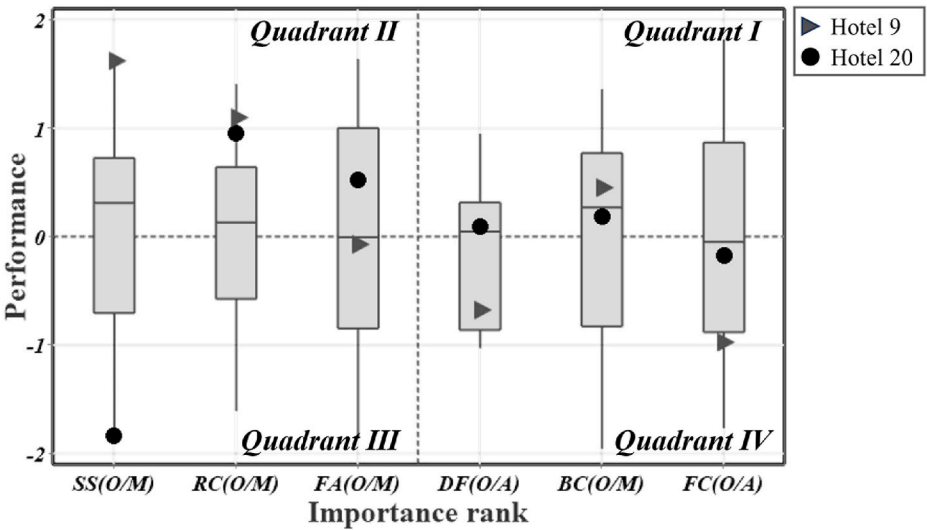
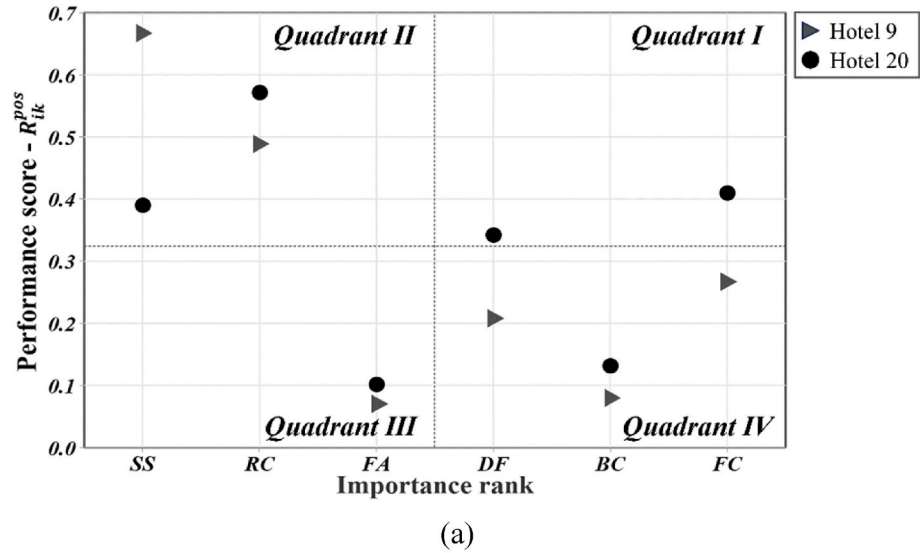
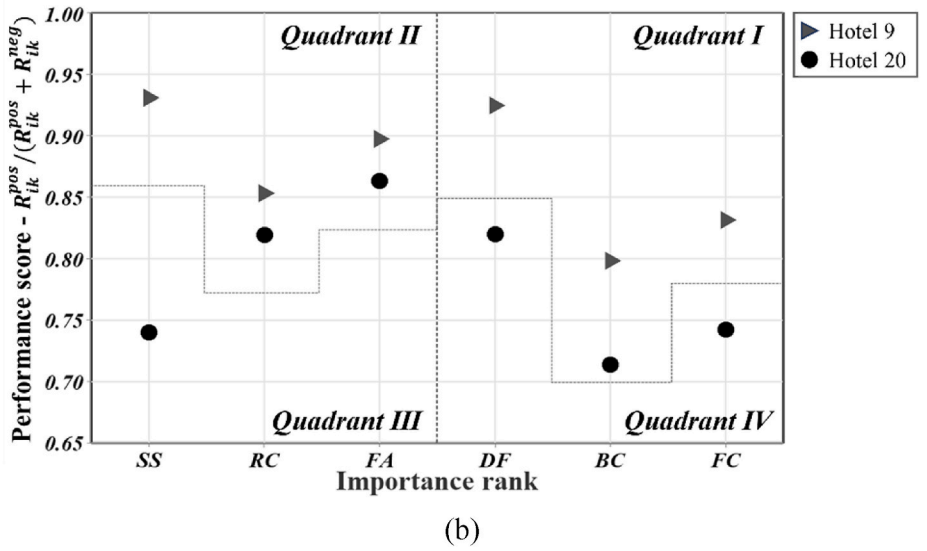


Fig. 5. KWIPCA matrix for 22 hotels (focusing on Hotels 9 and 20).



(a)



(b)

Fig. 6. Benchmark IPA matrices for Hotels 9 and 20: (a) Traditional IPA and (b) Wang et al. (2020).

Table 8
Quadrant positions of attributes in IPA matrices.

Attribute	Hotel 9			Hotel 20		
	KWIPCA (this study)	Traditional IPA	Wang et al. (2020)	KWIPCA (this study)	Traditional IPA	Wang et al. (2020)
SS	II	II	II	III	II	III
RC	II	II	II	II	II	II
FA	III	III	II	II	III	II
DF	IV	IV	I	I	I	IV
BC	I	IV	I	I	IV	I
FC	IV	IV	I	IV	I	IV

number of positive reviews is particularly critical, which lowers their position in the matrix.

Conversely, the matrix from Wang et al. (2020) emphasizes the positive-to-negative review ratio rather than absolute sentiment proportions. Consequently, even though Hotel 9 has a lower absolute proportion of positive reviews than its competitors, this matrix concludes that the performance of DF and FC is above average. However, this conclusion may not fully align with market perceptions. In such cases, the traditional IPA, which focuses on absolute sentiment proportions, may offer a more widely accepted interpretation.

As demonstrated in this case study, existing performance measures may lead to inconsistent conclusions in IPA. This inconsistency arises from a lack of consideration for market competitors and the Kano characteristics of attributes. The proposed KWIPCA offers more robust and widely acceptable conclusions by integrating these factors into the performance measure.

6. Conclusion

This study introduces the KWIPCA framework, a novel approach to understanding product quality and competitiveness through online review analytics by integrating Importance Performance Competitor Analysis (IPCA) with the Kano model. The KWIPCA matrix provides a robust and intuitive visualization of product quality and competitiveness by considering both the market importance of product attributes and their performance relative to all competitors. Its unique Kano-weighted performance measure enables a robust and comprehensive understanding of customer satisfaction, balancing positive and negative sentiments according to each attribute's Kano type. This approach allows for a strategic focus on key areas for improvement, offering clearer guidance for enhancing customer satisfaction and market positioning.

The case study, which analyzed reviews of five-star hotels, demonstrates the effectiveness of the KWIPCA framework in identifying strengths and weaknesses across various hotels. However, given its focus on a single industry (the hotel sector) and a specific geographical area (Seoul, Korea), the applicability of the sentiment analysis techniques and findings is inherently limited. Future research should broaden the application of the KWIPCA to further validate the framework, enhance its practical benefits, and provide richer, more diverse insights for improving product quality and market competitiveness across sectors.

Additionally, there is significant potential for enhancing the KWIPCA framework itself. While this study employed a lexicon-based review mining technique to identify sentiment, incorporating advanced machine learning approaches for contextual sentiment analysis, sentiment intensity scoring, and irony detection could significantly improve the accuracy and depth of sentiment evaluation. Furthermore, segmenting customer reviews based on factors such as demographics, region, or purpose of visit could yield a more detailed understanding of satisfaction, enabling businesses to develop more targeted improvement strategies.

Another promising direction for future research is to integrate the distinct features of physical products into the framework. The KWIPCA is indeed applicable to physical products, given that their reviews share structural similarities with service reviews. However, physical products

uniquely offer clear and quantitative specifications of their attributes. By incorporating these specifications alongside linguistic reviews, the KWIPCA framework could yield deeper insights into both performance and importance dimensions, ultimately enhancing its utility across a broader spectrum of products.

CRedit authorship contribution statement

Su-Ah Kim: Writing – original draft, Visualization, Validation, Software, Methodology. **Sohyun Park:** Methodology, Formal analysis, Data curation, Conceptualization. **Minjung Kwak:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Investigation, Formal analysis, Conceptualization. **Changmuk Kang:** Writing – review & editing, Validation, Supervision, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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